

Lecture 6 – 25 September

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Recap:

Genome sequence assembly and mapping

- Assembly

```
T A A
  A A T
    A T G
      T G C
        G C C
          C C A
            C A T
              A T G
                T G G
                  G G A
                    G A T
                      A T G
                        T A A T G C C A T G G A T G
```

- Mapping

```
T A A T G C C A T G G A T G
T A A       C C A       G A T
          G C C           A T G
                C C A
```

Gene expression count: 3

Data cleaning

1. Denoising (if applicable)
2. Remove outliers
3. Handle missing data
4. Remove duplicates
5. Encoding categorical data
6. Normalization

Data exploration

- Summary statistics
 - Location (mean, median)
 - Spread (range, percentiles, variance)
 - Frequency (mode)
- Visualization
 - histograms
 - box plots

Clustering

Why? → better **organization**, faster searching

Cluster items:

e.g. supermarket aisles, online shopping catalogues

Cluster people:

- Patients – diff category, diff treatment (e.g. children, adults, elders)
- Customers – optimized according to individual needs (personalization)

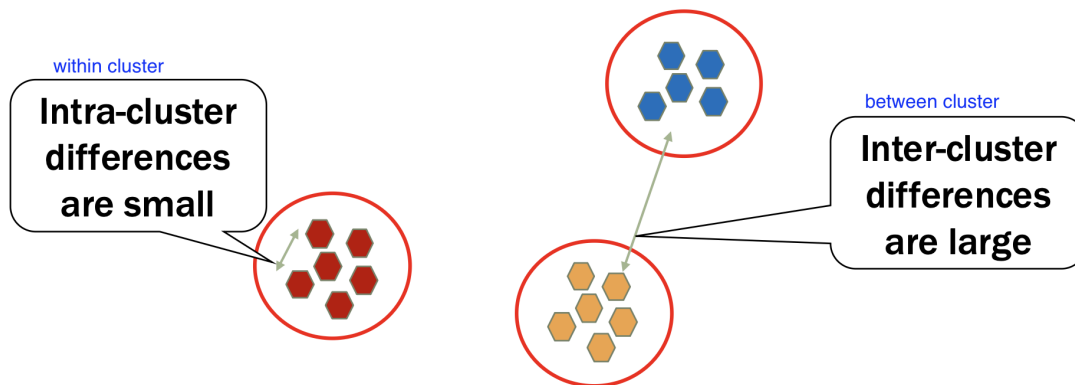
In biology:

- Genes:
 - Identify **coexpression**
 - Identify **differing expression** (e.g. disease marker, allowing early screening)

- Samples/cells:
 - Identify newer and/or existing disease **subtypes**
 - Identify **new cell types** within sample

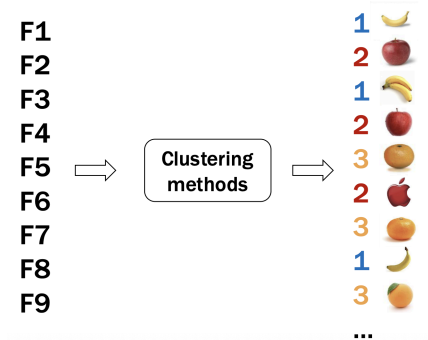
Clustering analysis

Finding groups where objects **within the group (intra-cluster)** are **similar** to each other, while more **different** with objects **outside the group (inter-cluster)**



- Clearer insight towards data distribution
- **Pre-processing** for later algorithms
 - e.g. similar types of documents, genes/proteins with similar morphology or functionality or base size
- **Reduce size** of large datasets
 - e.g. take avg of specific data item in a group → clustered form is simply representative info of the group → less size
- Preserve **privacy**
 - e.g. medical data (instead of per-person data, clustered data removes possible personal identification and only keeps representative data to be grouped)

- Example: grouping fruits
 1. Give fruit images to computer
 2. Images are changed to data, assign arbitrary names for them (e.g. F1, F2...)
 3. Data + clustering methods → clustering indicator
 4. Check correctness by comparing with their actual images



- Clustering necessities
 - **Data** (to be clustered)
 - **Similarity quantitative** measurement (defined differences → identify groups)
 - Clustering **algorithm** (clear running procedure (logic + steps) → reproducible results, reliable)

Similarity

- How **alike** two objects are
- Numerical, range [0,1]
- Higher when more alike

Dissimilarity

- How **different** two objects are
- Numerical, range [0, varied]
- Lower when more alike

Cosine similarity

$$\text{cos}(d_1, d_2) = \frac{d_1 \cdot d_2}{(|d_1| * |d_2|)}$$

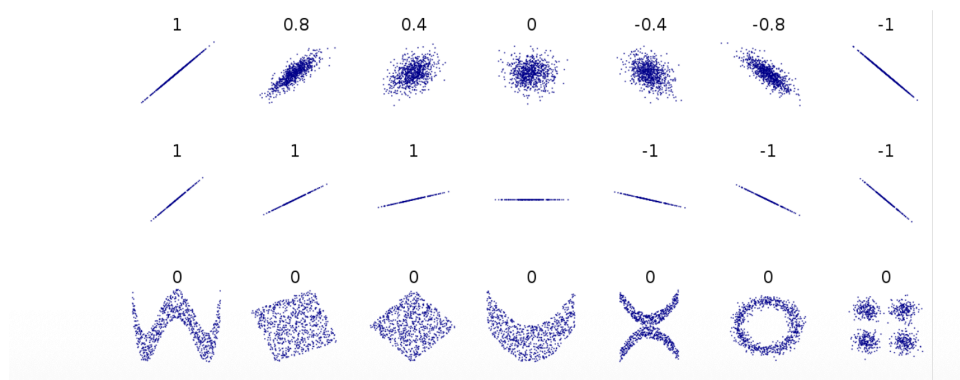
where d_1 and d_2 are vectors, $\cdot$ indicates dot product, and $|d|$ is the length of vector d

Correlation

Measures **linear relationship**

Uses Pearson's correlation coefficient

$$\rho_{X,Y} = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$



		2 Subtract Mean		3 Calculate ab, a ² and b ²		
Temp °C	Sales	"a"	"b"	a×b	a ²	b ²
14.2	\$215	-4.5	-\$187	842	20.3	34,969
16.4	\$325	-2.3	-\$77	177	5.3	5,929
11.9	\$185	-6.8	-\$217	1,476	46.2	47,089
15.2	\$332	-3.5	-\$70	245	12.3	4,900
18.5	\$406	-0.2	\$4	-1	0.0	16
22.1	\$522	3.4	\$120	408	11.6	14,400
19.4	\$412	0.7	\$10	7	0.5	100
25.1	\$614	6.4	\$212	1,357	41.0	44,944
23.4	\$544	4.7	\$142	667	22.1	20,164
18.1	\$421	-0.6	\$19	-11	0.4	361
22.6	\$445	3.9	\$43	168	15.2	1,849
17.2	\$408	-1.5	\$6	-9	2.3	36
18.7	\$402			5,325	177.0	174,757

1 Calculate Means

4 Sum Up

5 $\frac{5,325}{\sqrt{177.0 \times 174,757}} = 0.9575$

Euclidean distance

Identifies **shortest distance** between 2 points

Must normalize if scales/units in data differ (e.g. dataset with height (m) and weight (kg))

$$Ed(p, q) = \sqrt{\sum_{k=1}^m (p_k - q_k)^2}$$

where m is the number of dimensions, and p_k and q_k are values at k

Minkowski distance

Generalization of Euclidean distance

$$dist(p, q) = \left(\sum_{k=1}^m |p_k - q_k|^r \right)^{\frac{1}{r}}$$

Euclidean: $r = 2$ (blue)

$r = 1 \rightarrow$ manhattan/city block/taxicab distance (red)

$r = \infty \rightarrow$ supremum (max distance between 2 points)



All images and diagrams are taken from BMEG3105 Lecture 5 Slides unless stated otherwise.